

Section : Chemistry

Q.1 What is the source of nitrogen in Gabriel synthesis of primary aliphatic amines?

1. Sodium azide
2. Sodium nitrile
3. Potassium cyanide
4. Potassium phthalimide

Options 1. 1

2. 2
3. 3
4. 4

Q.2 Arrange the following compounds in increasing order of their acidity.

- (A) Phenol
- (B) Ethanol
- (C) Benzoic acid
- (D) p-Nitrobenzoic acid

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (A), (B), (D), (C)
3. (B), (A), (C), (D)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.3 The aqueous solutions of ZnSO_4 , AgNO_3 , FeCl_3 and WCl_6 are electrolysed for 2 hours using a current of 5 amperes. The decreasing order of mass deposited of the following metals at the cathode will be

- (A) Zn
- (B) Ag
- (C) Fe
- (D) W

(Atomic masses: $\text{Fe} = 56 \text{ g mol}^{-1}$, $\text{Zn} = 65 \text{ g mol}^{-1}$,
 $\text{Ag} = 108 \text{ g mol}^{-1}$, $\text{W} = 184 \text{ g mol}^{-1}$)

Choose the correct answer from the options given below:

- (1) (A), (B), (C), (D)
- (2) (A), (B), (D), (C)
- (3) (B), (A), (D), (C)
- (4) (C), (D), (B), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.4 Which one is a water-soluble protein?

1. Myosin
2. Albumin
3. Keratin
4. Collagen

Options 1. 1

2. 2
3. 3
4. 4

Q.5 In which of one the following all the concentration terms are independent of temperature?

1. molarity, ppm, molality
2. mass fraction, molarity, molality
3. mole fraction, mass fraction, molarity
4. mole fraction, mass fraction, molality

Options 1. 1

2. 2

3. 3

4. 4

Q.6 Match List-I with List-II :

List-I Molecule	List-II Van't Hoff Factor
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(A) Glucose	(I) 3
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(B) K_2SO_4	(II) 5
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(C) $Ca_3(PO_4)_2$	(III) 1
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(D) $K_3[Fe(CN)_6]$	(IV) 4
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Choose the correct answer from the options given below :

(1) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(2) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)

(3) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(4) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

Options 1. 1

2. 2

3. 3

4. 4

Q.7 Which of the following statement is not correct for Butan-2-ol?

1. It exists as a pair of non-superimposable mirror images
2. It is optically inactive
3. It is chiral
4. It has one stereocentre

Options 1. 1

2. 2

3. 3

4. 4

Q.8 When the electronic configuration of a transition metal element X in +3 oxidation state is $[Ar] 3d^5$, what is its atomic number ?

(1) 25

(2) 26

(3) 24

(4) 27

Options 1. 1

2. 2

3. 3

4. 4

Q.9 What do you understand by termolecular reactions

1. One molecule dissociating
2. Three molecules colliding
3. Two molecules colliding
4. Four molecules colliding

Options 1. 1

2. 2

3. 3

4. 4

Q.10 Which is the correct IUPAC name of the compound $\text{CH}_3\text{CH}=\text{CHC}(\text{Br})(\text{CH}_3)_2$?

(1) 4-Bromo-4-methylpent-2-ene

(2) 2-Bromo-2-methylpent-2-ene

(3) 4-Bromo-2-methylpent-2-ene

(4) 2-Bromo-4-methylpent-2-ene

Options 1. 1

2. 2

3. 3

4. 4

Q.11 Derive the electronic configuration of Ce^{3+} ion by using Hund's rule.

(1) $[\text{Xe}] 4f^1 5d^1 6s^2$

(2) $[\text{Xe}] 4f^1$

(3) $[\text{Rn}] 5f^2 6d^1 7s^2$

(4) $[\text{Rn}] 5f^1$

Options 1. 1

2. 2

3. 3

4. 4

Q.12 The C-O-H bond angle in alcohols is slightly less than tetrahedral angle $109^\circ 28'$ due to

1. bond pair - lone pair repulsion

2. lone pair - lone pair repulsion

3. bond pair - bond pair repulsion

4. Electronegativity of oxygen atom

Options 1. 1

2. 2

3. 3

4. 4

Q.13 Which of the following sugars does not undergo hydrolysis?

(A) Glucose

(B) Fructose

(C) Cane Sugar

(D) Lactose

Choose the correct answer from the options given below:

1. (A), (B) and (D) only

2. (A) and (B) only

3. (A), (B), (C) and (D)

4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.14 Arrange the following compounds in decreasing order of their acidity.

(A) Acetic acid

(B) Chloroacetic acid

(C) Dichloroacetic acid

(D) Trichloroacetic acid

Choose the correct answer from the options given below:

1. (D), (C), (B), (A)

2. (A), (B), (C) (D)
3. (A), (C), (B), (D)
4. (D), (B), (C), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.15 The quantity k in a rate law expression

1. is independent of temperature
2. is called the Arrhenius constant
3. is dimensionless
4. is independent of concentration

Options 1. 1

2. 2
3. 3
4. 4

Q.16 The standard reduction potential of Ag^+/Ag electrode is $+0.80\text{ V}$.

Choose the correct statement based on this information.

- (1) Ag^+/Ag couple is a stronger reducing agent than H^+/H_2 couple
(2) Ag^+/Ag couple is a weaker reducing agent than H^+/H_2 couple
(3) Ag can be easily oxidised as compared to H_2
(4) Ag^+/Ag will act as an anode when coupled with Standard Hydrogen Electrode to form a cell

Options 1. 1

2. 2
3. 3
4. 4

Q.17 Methyl chloride on treatment with excess of ammonia yields

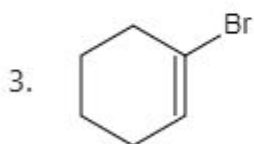
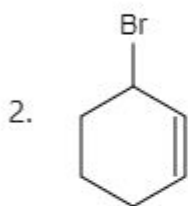
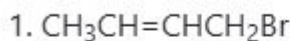
1. Methylamine
2. Dimethylamine
3. Trimethylamine
4. Mixture of methylamine, dimethylamine and trimethylamine

Options 1. 1

2. 2
3. 3
4. 4

Q.18

The compound representing a vinylic bromide is



Options 1. 1

2. 2

3. 3

4. 4

Q.19 Which of the following statements about the rate of a chemical reaction are correct?

(A) The rate of reaction is always proportional to the concentration of reactants

(B) An exponential time course is followed by the first order reaction

(C) The unit of rate of reaction in the gaseous state is atm/s

(D) The concentration of reactant decreases exponentially with time for a zero order reaction

Choose the correct answer from the options given below:

1. (A) and (D) only

2. (B), (C) and (D) only

3. (A), (B), (C) and (D)

4. (B) and (C) only

Options 1. 1

2. 2

3. 3

4. 4

Q.20 What types of bond (linkage) are present in nucleotide?

1. Peptide linkage

2. Ionic bond

3. Phosphodiester linkage

4. O-glycosidic linkage

Options 1. 1

2. 2

3. 3

4. 4

Q.21 Match List-I with List-II :

List-I

Representation

List-II

Common name

(A) 10^{-6} M solution (I) Decimolar solution

(B) 0.1 M solution (II) Micromolar solution

- (C) 0.001 M solution (III) Millimolar solution
(D) 0.01 M solution (IV) Centimolar solution

Choose the correct answer from the options given below :

- (1) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)
(2) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)
(3) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)
(4) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

Options 1. 1

2. 2
3. 3
4. 4

Q.22 Write down the number of 3d electrons in Co^{2+} ion.

- (1) 7
(2) 5
(3) 6
(4) 2

Options 1. 1

2. 2
3. 3
4. 4

Q.23 Internal acid base reaction of amino acids is due to formation of

1. Zwitter ion
2. A lactum
3. An amide
4. A peptide

Options 1. 1

2. 2
3. 3
4. 4

Q.24 Match List-I with List-II

List-I
(Electronic configuration)

List-II

(Ion)

- | | |
|-------------------------------|------------------------|
| (A) $[\text{Ar}] 3d^{10}$ | (I) Ce^{4+} |
| (B) $[\text{Xe}]$ | (II) Cu^+ |
| (C) $[\text{Rn}]$ | (III) Th^{4+} |
| (D) $[\text{Xe}] 4f^{14}5d^1$ | (IV) Lu^{2+} |

Choose the correct answer from the options given below :

- (1) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)
(2) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)
(3) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)
(4) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

Options 1. 1

2. 2
3. 3
4. 4

Q.25 Reduction of nitrobenzene by which of the following reagent gives aniline?

- (A) Sn/HCl
- (B) Fe/HCl
- (C) $\text{H}_2 - \text{Pd}$
- (D) $\text{Sn}/\text{NH}_4\text{OH}$

Choose the correct answer from the options given below:

- (1) (A), (B) and (D) only
- (2) (A), (B) and (C) only
- (3) (A), (B), (C) and (D)
- (4) (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.26 Which sodium halide is employed in Finkelstein reaction?

- 1. Sodium chloride
- 2. Sodium bromide
- 3. Sodium fluoride
- 4. Sodium iodide

Options 1. 1

2. 2

3. 3

4. 4

Q.27 What is the total number of chiral carbons present in glucopyranose and fructofuranose respectively?

- 1. 3 and 4
- 2. 5 and 4
- 3. 5 and 3
- 4. 4 and 5

Options 1. 1

2. 2

3. 3

4. 4

Q.28 Sucrose on hydrolysis gives

- 1. glucose and fructose
- 2. glucose and galactose
- 3. maltose and fructose
- 4. galactose and lactose

Options 1. 1

2. 2

3. 3

4. 4

Q.29 The half-cell reaction is the one that

- 1. consumes half mole of an electrolyte
- 2. takes place at one electrode
- 3. reduces the concentration of electrolyte to half of its initial value
- 4. reduces the electrode potential to half its initial value

Options 1. 1

2. 2

3. 3

4. 4

Q.30 Calculate the magnetic moment of $\text{K}_4[\text{Mn}(\text{CN})_6]$ complex.

- (1) 5.92 BM

- (2) 0 BM
(3) 1.73 BM
(4) 2.84 BM

Options 1. 1

2. 2
3. 3
4. 4

Q.31 Toluene reacts with a halogen in the presence of FeCl_3 , giving a mixture of ortho and para halo compounds. This type of reaction is called

1. Nucleophilic substitution reaction
2. Electrophilic substitution reaction
3. Free radical addition reaction
4. Nucleophilic addition reaction

Options 1. 1

2. 2
3. 3
4. 4

Q.32 From the given standard reduction potentials, arrange the metals in decreasing order of their reducing power.

(A) $E_{\text{Zn}^{2+}/\text{Zn}}^0 = -0.76 \text{ V}$

(B) $E_{\text{Mg}^{2+}/\text{Mg}}^0 = -2.37 \text{ V}$

(C) $E_{\text{Al}^{3+}/\text{Al}}^0 = -1.67 \text{ V}$

(D) $E_{\text{Ag}^+/\text{Ag}}^0 = +0.80 \text{ V}$

Choose the correct answer from the options given below:

- (1) (A), (B), (C), (D)
(2) (D), (A), (C), (B)
(3) (B), (A), (D), (C)
(4) (B), (C), (A), (D)

Options 1. 1

2. 2
3. 3
4. 4

Q.33 The rate determining step for a complex reaction is

1. the slowest step
2. the fastest step
3. the first step
4. the step involving the major reactant

Options 1. 1

2. 2
3. 3
4. 4

Q.34 The reagents used for dehydration of alcohols are:

- (A) Concentrated H_2SO_4
(B) Phosphoric acid
(C) Alumina
(D) Anhydrous ZnCl_2

Choose the correct answer from the options given below:

- (1) (A), (B) and (D) only
(2) (A), (B) and (C) only

(3) (A), (B), (C) and (D)

(4) (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.35 The following statements are based on the osmotic pressure of solutions. The correct statements are

(A) Molar concentration is same for isotonic solutions

(B) Solutions which have same osmotic pressure at same temperature are called isotonic solutions

(C) No osmosis occurs when isotonic solutions are separated by semi-permeable membrane

(D) A 1.9% Sodium Chloride solution has equivalent osmotic pressure to that of fluid inside blood cells

Choose the correct answer from the options given below:

1. (A) and (D) only

2. (A), (B) and (C) only

3. (A), (C) and (D) only

4. (B) and (C) only

Options 1. 1

2. 2

3. 3

4. 4

Q.36 Arrange the following in the increasing order of their solubility in n-octane on the basis of their solute-solvent interaction:

(A) KCl

(B) CH₃OH

(C) CH₃CN

(D) Cyclohexane

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)

2. (A), (C), (D), (B)

3. (B), (A), (C), (D)

4. (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.37 Match List-I with List-II

List-I

Rate expression

List-II

Overall order of reaction

(A) $r = k[A]^{0.2}[B]^{0.3}[C]$ (I) $\frac{3}{10}$

(B) $r = k[A]^{0.2}[B]^{-0.3}[C]$ (II) 1

(C) $r = k[A][B]^{0.3}[C]^{-1}$ (III) $\frac{3}{2}$

(D) $r = k[A]^{0.2}[B]^{0.3}[C]^{0.5}$ (IV) $\frac{9}{10}$

Choose the correct answer from the options given below:

(1) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

(2) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(3) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)

(4) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

Options 1. 1

2. 2
3. 3
4. 4

Q.38 Which of the following compounds will not undergo aldol condensation?

- (1) Formaldehyde
- (2) Ethyl methyl ketone
- (3) Acetaldehyde
- (4) Propanal

Options 1. 1

2. 2
3. 3
4. 4

Q.39 The correct SI unit of molar conductivity is

- (1) $\text{S cm}^{-1} \text{ mol}^{-1} \text{ L}$
- (2) $\text{S cm}^2 \text{ mol}^{-1}$
- (3) $\text{S m}^2 \text{ mol}^{-1}$
- (4) $\text{S m}^2 \text{ mol}$

Options 1. 1

2. 2
3. 3
4. 4

Q.40 Match List-I with List-II

List-I	List-II
Test	Identification

- | | |
|---------------------|--|
| (A) Hinsberg's Test | (I) Primary, Secondary and tertiary Amines |
| (B) Isocyanide test | (II) Aliphatic and Aromatic primary amines |
| (C) Azo dye test | (III) Primary Aromatic amines |
| (D) Tollen's Test | (IV) Reducing sugars |

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.41 Read the Passage carefully and answer the questions

When a ligand is bound to a metal ion through a single donor atom, it is said to be a unidentate ligand but when it binds through two donor atoms, then it is said to be a didentate ligand and when several donor atoms are present in a single ligand, it is said to be a polydentate ligand. When a di- or polydentate ligand uses its two or more donor atoms simultaneously to bind a single metal ion, it is said to be a chelate ligand. The number of such ligating groups is called the denticity of the ligand. A ligand which has two different donor atoms and either of the two ligates in the complex is called an ambidentate ligand. The number of ligand donor atoms to which the metal is directly bonded, is called the coordination number of the metal ion in the complex. Complexes in which a metal is bound to only one kind of donor groups are known as homoleptic and in which a metal is bound to more than one kind of donor groups are known as heteroleptic complexes.

Which of the following molecule/ion is not expected to behave as a ligand?

- (1) NO
- (2) NH_4^+
- (3) CN^-
- (4) I^-

Options 1. 1

2. 2
3. 3
4. 4

Q.42 Read the Passage carefully and answer the questions

When a ligand is bound to a metal ion through a single donor atom, it is said to be a unidentate ligand but when it binds through two donor atoms, then it is said to be a didentate ligand and when several donor atoms are present in a single ligand, it is said to be a polydentate ligand. When a di- or polydentate ligand uses its two or more donor atoms simultaneously to bind a single metal ion, it is said to be a chelate ligand. The number of such ligating groups is called the denticity of the ligand. A ligand which has two different donor atoms and either of the two ligates in the complex is called ambidentate ligands. The number of ligand donor atoms to which the metal is directly bonded, is called coordination number of the metal ion in the complex. Complexes in which a metal is bound to only one kind of donor groups are known as homoleptic and in which a metal is bound to more than one kind of donor groups are known as heteroleptic complexes.

Which of the following is a chelate complex?

- (1) $[\text{Co}(\text{en})_3]\text{Cl}_3$
- (2) $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)](\text{NO}_3)_2$
- (3) $[\text{Pt}(\text{NH}_3)(\text{H}_2\text{O})\text{Cl}_2]$
- (4) $[\text{Fe}(\text{NH}_3)_2(\text{CN})_4]\text{Cl}$

Options 1. 1

2. 2
3. 3
4. 4

Q.43 Read the Passage carefully and answer the questions

When a ligand is bound to a metal ion through a single donor atom, it is said to be a unidentate ligand but when it binds through two donor atoms, then it is said to be a didentate ligand and when several donor atoms are present in a single ligand, it is said to be a polydentate ligand. When a di- or polydentate ligand uses its two or more donor atoms simultaneously to bind a single metal ion, it is said to be a chelate ligand. The number of such ligating groups is called the denticity of the ligand. A ligand which has two different donor atoms and either of the two ligates in the complex is called ambidentate ligands. The number of ligand donor atoms to which the metal is directly bonded, is called coordination number of the metal ion in the complex. Complexes in which a metal is bound to only one kind of donor groups are known as homoleptic and in which a metal is bound to more than one kind of donor groups are known as heteroleptic complexes.

The coordination number of Co and Al in $[\text{CoCl}(\text{en})_2]\text{Cl}$ and $\text{K}_2[\text{Al}(\text{C}_2\text{O}_4)_3]$, respectively, are

- (1) 3 and 3
- (2) 6 and 6
- (3) 3 and 6
- (4) 5 and 6

Options 1. 1

2. 2
3. 3
4. 4

Q.44 Read the Passage carefully and answer the questions

When a ligand is bound to a metal ion through a single donor atom, it is said to be a unidentate ligand but when it binds through two donor atoms, then it is said to be a didentate ligand and when several donor atoms are

present in a single ligand, it is said to be a polydentate ligand. When a di- or polydentate ligand uses its two or more donor atoms simultaneously to bind a single metal ion, it is said to be a chelate ligand. The number of such ligating groups is called the denticity of the ligand. ligand which has two different donor atoms and either of the two ligates in the complex is called ambidentate ligands. The number of ligand donor atoms to which the metal is directly bonded, is called coordination number of the metal ion in the complex. Complexes in which a metal is bound to only one kind of donor groups are known as homoleptic and in which a metal is bound to more than one kind of donor groups are known as heteroleptic complexes.

Which among the following is an ambidentate ligand?

- (1) SCN^-
- (2) NH_3
- (3) H_2O
- (4) CO

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.45 Read the Passage carefully and answer the questions

When a ligand is bound to a metal ion through a single donor atom, it is said to be a unidentate ligand but when it bind through two donor atoms, then it is said to be didentate ligand and when several donor atoms are present in a single ligand, it is said to be a polydentate ligand. When a di- or polydentate ligand uses its two or more donor atoms simultaneously to bind a single metal ion, it is said to be a chelate ligand. The number of such ligating groups is called the denticity of the ligand. ligand which has two different donor atoms and either of the two ligates in the complex is called ambidentate ligands. The number of ligand donor atoms to which the metal is directly bonded, is called coordination number of the metal ion in the complex. Complexes in which a metal is bound to only one kind of donor groups are known as homoleptic and in which a metal is bound to more than one kind of donor groups are known as heteroleptic complexes.

To make an octahedral complex of Mg^{2+} ion, how many molecules of EDTA are used?

- (1) one
- (2) six
- (3) three
- (4) two

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.46 Read the passage carefully and answer the questions

The carbonyl group of aldehydes and ketones is highly polar. The positively charged carbon is readily attacked by electron-rich nucleophile and the negatively charged oxygen atom is attacked by an electro-deficient electrophile. Aldehydes and ketones undergo nucleophilic addition reactions in the presence of acid or a base. The mechanism involves attack of a nucleophile on the electron deficient carbonyl carbon to form a tetrahedral intermediate. A new bond is formed between the carbon and oxygen. Protonation of the anion furnished by the reaction medium yields a neutral addition product. The addition product is same whether the reaction is catalyzed by acid or a base. For example in the presence of base aldehydes and ketones react with hydrogen cyanide to form cyanohydrin.

Which of the following reactions is not a nucleophilic addition reaction of aldehydes and ketones?

- 1. Addition of water
- 2. Addition of sodium Hydrogen Sulphite
- 3. Addition of alcohol
- 4. Addition of sodium borohydride

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.47 Read the passage carefully and answer the questions

The carbonyl group of aldehydes and ketones is highly polar. The positively charged carbon is readily attacked by electron-rich nucleophile and the negatively charged oxygen atom is attacked by an electro-deficient electrophile. Aldehydes and ketones undergo nucleophilic addition reactions in the presence of acid or a base. The mechanism involves attack of a nucleophile on the electron deficient carbonyl carbon to form a tetrahedral intermediate. A new bond is formed between the carbon and oxygen. Protonation of the anion furnished by the reaction medium yields a neutral addition product. The addition product is same whether the reaction is catalyzed by acid or a base. For example in the presence of base aldehydes and ketones react with hydrogen cyanide to form cyanohydrin.

Which of the following compounds will undergo nucleophilic addition readily?

- 1. Formaldehyde
- 2. Acetaldehyde
- 3. Benzaldehyde
- 4. Butanone

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.48 Read the passage carefully and answer the questions

The carbonyl group of aldehydes and ketones is highly polar. The positively charged carbon is readily attacked by electron-rich nucleophile and the negatively charged oxygen atom is attacked by an electro-deficient electrophile. Aldehydes and ketones undergo nucleophilic addition reactions in the presence of acid or a base. The mechanism involves attack of a nucleophile on the electron deficient carbonyl carbon to form a tetrahedral intermediate. A new bond is formed between the carbon and oxygen. Protonation of the anion furnished by the reaction medium yields a neutral addition product. The addition product is same whether the reaction is catalyzed by acid or a base. For example in the presence of base aldehydes and ketones react with hydrogen cyanide to form cyanohydrin.

What is the product formed when an aldehyde other than formaldehyde is treated with Grignard reagent?

- 1. Carboxylic acid
- 2. Primary alcohol
- 3. Secondary alcohol
- 4. Tertiary alcohol

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.49 Read the passage carefully and answer the questions

The carbonyl group of aldehydes and ketones is highly polar. The positively charged carbon is readily attacked by electron-rich nucleophile and the negatively charged oxygen atom is attacked by an electro-deficient electrophile. Aldehydes and ketones undergo nucleophilic addition reactions in the presence of acid or a base. The mechanism involves attack of a nucleophile on the electron deficient carbonyl carbon to form a tetrahedral intermediate. A new bond is formed between the carbon and oxygen. Protonation of the anion furnished by the reaction medium yields a neutral addition product. The addition product is same whether the reaction is catalyzed by acid or a base. For example in the presence of base aldehydes and ketones react with hydrogen cyanide to form cyanohydrin.

The source of nucleophile in the base catalyzed cyanohydrin formation is

1. HCN
2. KCN
3. NaCN
4. CH₃CN

Options 1. 1

2. 2
3. 3
4. 4

Q.50 Read the passage carefully and answer the questions

The carbonyl group of aldehydes and ketones is highly polar. The positively charged carbon is readily attacked by electron-rich nucleophile and the negatively charged oxygen atom is attacked by an electro-deficient electrophile. Aldehydes and ketones undergo nucleophilic addition reactions in the presence of acid or a base. The mechanism involves attack of a nucleophile on the electron deficient carbonyl carbon to form a tetrahedral intermediate. A new bond is formed between the carbon and oxygen. Protonation of the anion furnished by the reaction medium yields a neutral addition product. The addition product is same whether the reaction is catalyzed by acid or a base. For example in the presence of base aldehydes and ketones react with hydrogen cyanide to form cyanohydrin.

During nucleophilic addition reaction the carbonyl carbon undergoes a change in hybridization from

1. sp² to sp³
2. sp³ to sp²
3. sp to sp²
4. sp² to sp

Options 1. 1

2. 2
3. 3
4. 4